Run report: run-20261018T1605Z

Environment and command

```
run: 20261018T1605Z (Parts G and H)
protocol commit: d5bcdc081 (freeze 095c45c8a + amendment 1)
rustc 1.97.0 (2d8144b78 2026-07-07)
host: Linux 6.18.44-fc-v139 x86_64, 4 cores, Intel(R) Xeon(R) Processor @ 2.80GHz, 15 GB
command: rustc --edition 2021 -O examples/weil_restriction_isogeny_census.rs -o wric && ./wric research/
weil_restriction_isogeny_census_20261018/results/run-20261018T1605Z G H
tool sha256: 3a87f373cae06cd24cd5f80950a7017a26a4087bdc56cd99ffa9a96689645d8c (rustfmt-formatted; the committed file)
threads: 1
smoke runs before this run (not evidence, outputs discarded): one Part G run (75 s, checks 1-4 as here) and one Part
H run at q = 5, 7 that exposed a wrong expected degree in check 6 for the ℓ = 2 cubic (degree 3, not (ℓ^2 − 1)/2); the
check was corrected in the tool before this run. No other change.

note (after the run): the committed tool was changed after this run in two ways that touch no arithmetic: two
regression tests were added (the Jordan class at ℓ = 7 has 7 submodules of dimension 3, 3 kernels, 2 Tian kernels;
nonsplit classes have no submodule of dimension 3) and clippy lints were fixed (is_multiple_of, an iterator loop in
rref, type aliases). sha256 of the committed file: 04dead65a5847933f25609e29dcbea99349c8720d437a70613c93d118ba27c6c.
Verified: kernels.csv of a rerun is byte-identical to this run's (seconds column excluded), and the q = 5, 7 rows of
curves.csv and by_class.csv are byte-identical.
```

Tool log

```
part G cell n=2 l=2: 0.00s
part G cell n=2 l=3: 0.00s
part G cell n=2 l=5: 0.02s
part G cell n=2 l=7: 0.13s
part G cell n=3 l=2: 0.00s
part G cell n=3 l=3: 0.01s
part G cell n=3 l=5: 0.92s
part G cell n=3 l=7: 15.25s
part G cell n=4 l=2: 0.00s
part G cell n=4 l=3: 0.28s
part G cell n=4 l=5: 55.55s
check1 (n=3, l=2: Tian kernels only for phi = I): violations=0
check2 (n=2: every Tian kernel is a graph with det psi = -qbar^-1 and phi psi^2 = I): violations=0
check3 (P2: at (n,l,qbar)=(3,3,1) a class without eigenvalue 1 has a Tian kernel): classes=1
check4 (representative independence): mismatches=0
part H q=3: 56 curves, 0.0s
part H q=5: 252 curves, 0.2s
part H q=7: 690 curves, 0.2s
part H q=11: 2662 curves, 2.2s
part H q=13: 4400 curves, 5.6s
q=3 check5 classes=56 expected=56 ok
q=3 hasse_violations=0 class_not_found=0 check6_violations=0 check7_violations=0 check8_violations=0
q=3 check9 prime_order=8 prime_with_tian_some_odd_ell=0 all=56 all_with_tian_some_odd_ell=28
q=5 check5 classes=252 expected=252 ok
q=5 hasse_violations=0 class_not_found=0 check6_violations=0 check7_violations=0 check8_violations=0
q=5 check9 prime_order=18 prime_with_tian_some_odd_ell=0 all=252 all_with_tian_some_odd_ell=156
q=7 check5 classes=690 expected=690 ok
q=7 hasse_violations=0 class_not_found=0 check6_violations=0 check7_violations=0 check8_violations=0
q=7 check9 prime_order=61 prime_with_tian_some_odd_ell=24 all=690 all_with_tian_some_odd_ell=372
q=11 check5 classes=2662 expected=2662 ok
q=11 hasse_violations=0 class_not_found=0 check6_violations=0 check7_violations=0 check8_violations=0
q=11 check9 prime_order=150 prime_with_tian_some_odd_ell=57 all=2662 all_with_tian_some_odd_ell=1984
q=13 check5 classes=4400 expected=4400 ok
q=13 hasse_violations=0 class_not_found=0 check6_violations=0 check7_violations=0 check8_violations=0
q=13 check9 prime_order=323 prime_with_tian_some_odd_ell=144 all=4400 all_with_tian_some_odd_ell=2714
total 80.4s
```

Checks

```
check1 (n=3, l=2: Tian kernels only for phi = I): violations=0
check2 (n=2: every Tian kernel is a graph with det psi = -qbar^-1 and phi psi^2 = I): violations=0
check3 (P2: at (n,l,qbar)=(3,3,1) a class without eigenvalue 1 has a Tian kernel): classes=1
check4 (representative independence): mismatches=0
q=3 check5 classes=56 expected=56 ok
q=3 hasse_violations=0 class_not_found=0 check6_violations=0 check7_violations=0 check8_violations=0
q=3 check9 prime_order=8 prime_with_tian_some_odd_ell=0 all=56 all_with_tian_some_odd_ell=28
q=5 check5 classes=252 expected=252 ok
q=5 hasse_violations=0 class_not_found=0 check6_violations=0 check7_violations=0 check8_violations=0
q=5 check9 prime_order=18 prime_with_tian_some_odd_ell=0 all=252 all_with_tian_some_odd_ell=156
q=7 check5 classes=690 expected=690 ok
q=7 hasse_violations=0 class_not_found=0 check6_violations=0 check7_violations=0 check8_violations=0
q=7 check9 prime_order=61 prime_with_tian_some_odd_ell=24 all=690 all_with_tian_some_odd_ell=372
q=11 check5 classes=2662 expected=2662 ok
q=11 hasse_violations=0 class_not_found=0 check6_violations=0 check7_violations=0 check8_violations=0
q=11 check9 prime_order=150 prime_with_tian_some_odd_ell=57 all=2662 all_with_tian_some_odd_ell=1984
q=13 check5 classes=4400 expected=4400 ok
q=13 hasse_violations=0 class_not_found=0 check6_violations=0 check7_violations=0 check8_violations=0
q=13 check9 prime_order=323 prime_with_tian_some_odd_ell=144 all=4400 all_with_tian_some_odd_ell=2714
```

Part H summary (summary.csv)

q	classes	prime_order	cofactor_2	full_2_torsion	ell	with_tian	prime_with_tian	cof2_with_tian	prime_any_odd_ell	all_any_odd_ell
3	56	8	7	9	2	9	0	0	0	28
3	56	8	7	9	5	26	0	6	0	28
3	56	8	7	9	7	8	0	0	0	28
5	252	18	25	42	2	42	0	0	0	156
5	252	18	25	42	3	124	0	0	0	156
5	252	18	25	42	7	40	0	0	0	156
7	690	61	48	115	2	115	0	0	24	372
7	690	61	48	115	3	58	3	6	24	372
7	690	61	48	115	5	342	24	0	24	372
11	2662	150	150	443	2	443	0	0	57	1984
11	2662	150	150	443	3	1330	0	0	57	1984
11	2662	150	150	443	5	710	57	78	57	1984
11	2662	150	150	443	7	758	12	12	57	1984
13	4400	323	288	734	2	734	0	0	144	2714
13	4400	323	288	734	3	368	12	24	144	2714
13	4400	323	288	734	5	2198	135	114	144	2714
13	4400	323	288	734	7	732	0	0	144	2714

Part G, n = 3 (kernels.csv)

ell	qbar	class	tr	det	eigenvalue_one	submodules_dim_n	kernels	tian_kernels	types_d01
2	1	scalar	0	1	true	15	9	6	1:6
2	1	jordan	0	1	true	1	1	0	-
2	1	nonsplit	1	1	false	0	0	0	-
3	1	nonsplit	0	1	false	0	0	0	-
3	1	scalar	1	1	false	40	16	12	1:12
3	1	jordan	1	1	false	1	1	0	-
3	1	scalar	2	1	true	40	16	12	1:12
3	1	jordan	2	1	true	1	1	0	-
3	2	split	0	2	true	4	4	2	1:2
3	2	nonsplit	1	2	false	0	0	0	-
3	2	nonsplit	2	2	false	0	0	0	-
5	1	split	0	1	false	4	4	2	1:2
5	1	nonsplit	1	1	false	0	0	0	-
5	1	scalar	2	1	true	156	36	30	1:30
5	1	jordan	2	1	true	1	1	0	-
5	1	scalar	3	1	false	156	36	30	1:30
5	1	jordan	3	1	false	1	1	0	-
5	1	nonsplit	4	1	false	0	0	0	-
5	2	nonsplit	0	3	false	0	0	0	-
5	2	split	1	3	false	4	4	2	1:2
5	2	nonsplit	2	3	false	0	0	0	-
5	2	nonsplit	3	3	false	0	0	0	-
5	2	split	4	3	true	4	4	2	1:2
5	3	nonsplit	0	2	false	0	0	0	-
5	3	nonsplit	1	2	false	0	0	0	-
5	3	split	2	2	false	4	4	2	1:2
5	3	split	3	2	true	4	4	2	1:2
5	3	nonsplit	4	2	false	0	0	0	-
5	4	split	0	4	true	4	4	2	1:2
5	4	scalar	1	4	false	156	36	30	1:30

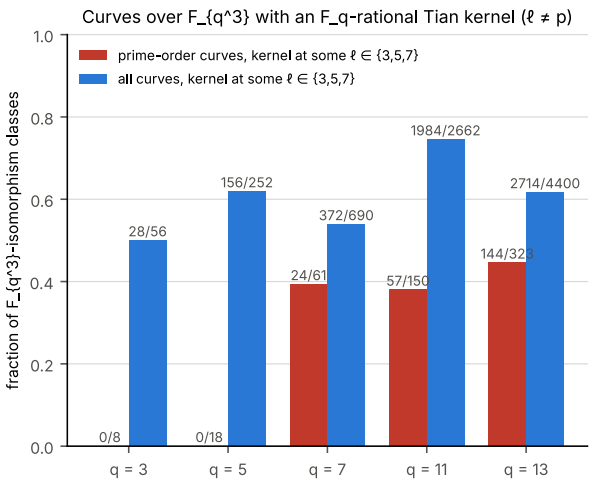
5	4	jordan	1	4	false	1	1	0	-
5	4	nonsplit	2	4	false	0	0	0	-
5	4	nonsplit	3	4	false	0	0	0	-
5	4	scalar	4	4	false	156	36	30	1:30
5	4	jordan	4	4	false	1	1	0	-
7	1	nonsplit	0	1	false	0	0	0	-
7	1	split	1	1	false	2	2	0	-
7	1	scalar	2	1	true	560	80	72	1:72
7	1	jordan	2	1	true	7	3	2	1:2
7	1	nonsplit	3	1	false	0	0	0	-
7	1	nonsplit	4	1	false	0	0	0	-
7	1	scalar	5	1	false	560	80	72	1:72
7	1	jordan	5	1	false	7	3	2	1:2
7	1	split	6	1	false	2	2	0	-
7	2	nonsplit	0	1	false	0	0	0	-
7	2	split	1	1	false	2	2	0	-
7	2	scalar	2	1	true	560	80	72	1:72
7	2	jordan	2	1	true	7	3	2	1:2
7	2	nonsplit	3	1	false	0	0	0	-
7	2	nonsplit	4	1	false	0	0	0	-
7	2	scalar	5	1	false	560	80	72	1:72
7	2	jordan	5	1	false	7	3	2	1:2
7	2	split	6	1	false	2	2	0	-
7	3	split	0	6	true	20	8	6	1:6
7	3	nonsplit	1	6	false	0	0	0	-
7	3	split	2	6	false	2	2	0	-
7	3	nonsplit	3	6	false	0	0	0	-
7	3	nonsplit	4	6	false	0	0	0	-
7	3	split	5	6	false	2	2	0	-
7	3	nonsplit	6	6	false	0	0	0	-
7	4	nonsplit	0	1	false	0	0	0	-
7	4	split	1	1	false	2	2	0	-
7	4	scalar	2	1	true	560	80	72	1:72
7	4	jordan	2	1	true	7	3	2	1:2
7	4	nonsplit	3	1	false	0	0	0	-
7	4	nonsplit	4	1	false	0	0	0	-
7	4	scalar	5	1	false	560	80	72	1:72
7	4	jordan	5	1	false	7	3	2	1:2
7	4	split	6	1	false	2	2	0	-
7	5	split	0	6	true	20	8	6	1:6
7	5	nonsplit	1	6	false	0	0	0	-
7	5	split	2	6	false	2	2	0	-
7	5	nonsplit	3	6	false	0	0	0	-
7	5	nonsplit	4	6	false	0	0	0	-
7	5	split	5	6	false	2	2	0	-
7	5	nonsplit	6	6	false	0	0	0	-
7	6	split	0	6	true	20	8	6	1:6
7	6	nonsplit	1	6	false	0	0	0	-
7	6	split	2	6	false	2	2	0	-
7	6	nonsplit	3	6	false	0	0	0	-
7	6	nonsplit	4	6	false	0	0	0	-
7	6	split	5	6	false	2	2	0	-

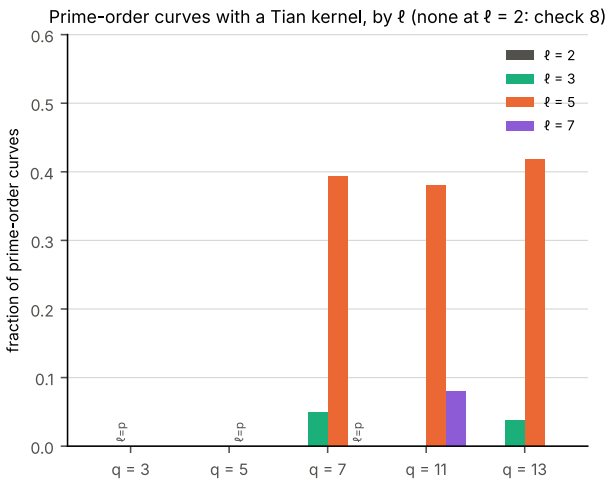
7	6	nonsplit	6	6	false	0	0	0	-

Part H: classes with a Tian kernel at odd ℓ (by_class.csv)

q	ell	qbar	class	tr	det	curves	prime_order_curves	cofactor_2_curves
3	5	3	split	2	2	13	0	6
3	5	3	split	3	2	13	0	0
3	7	3	split	0	6	8	0	0
5	3	2	split	0	2	124	0	0
5	7	5	split	0	6	40	0	0
7	3	1	scalar	1	1	29	3	6
7	3	1	scalar	2	1	29	0	0
7	5	2	split	1	3	171	24	0
7	5	2	split	4	3	171	0	0
11	3	2	split	0	2	1330	0	0
11	5	1	scalar	2	1	22	0	0
11	5	1	scalar	3	1	22	3	0
11	5	1	split	0	1	666	54	78
11	7	4	jordan	2	1	370	0	0
11	7	4	jordan	5	1	370	12	12
11	7	4	scalar	2	1	9	0	0
11	7	4	scalar	5	1	9	0	0
13	3	1	scalar	1	1	184	12	24
13	3	1	scalar	2	1	184	0	0
13	5	3	split	2	2	1099	135	114
13	5	3	split	3	2	1099	0	0
13	7	6	split	0	6	732	0	0

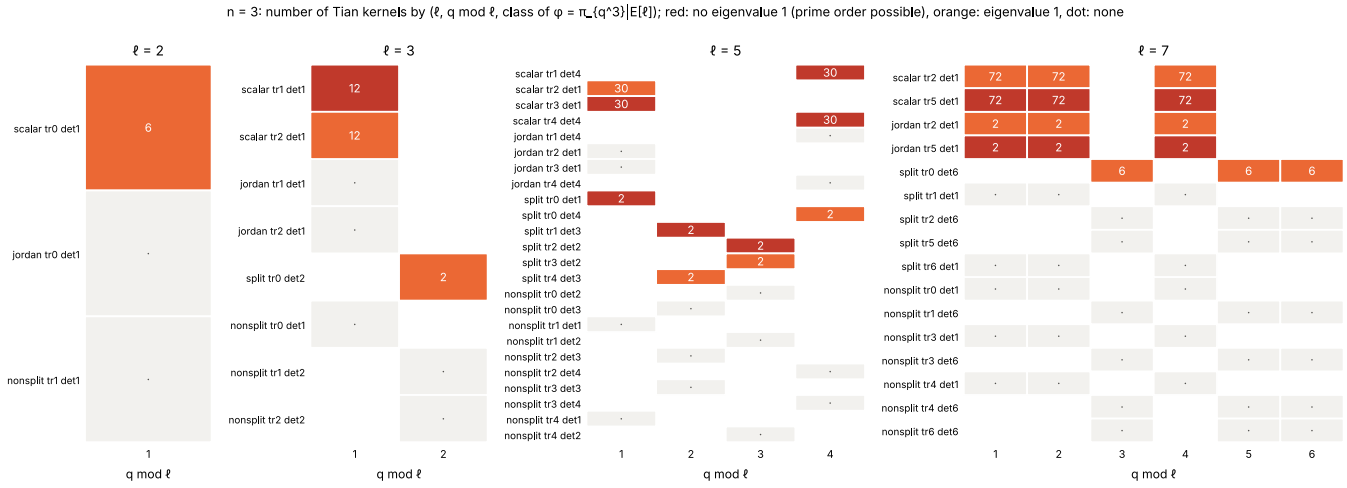
Graph: Curves with a Tian kernel by q and ℓ





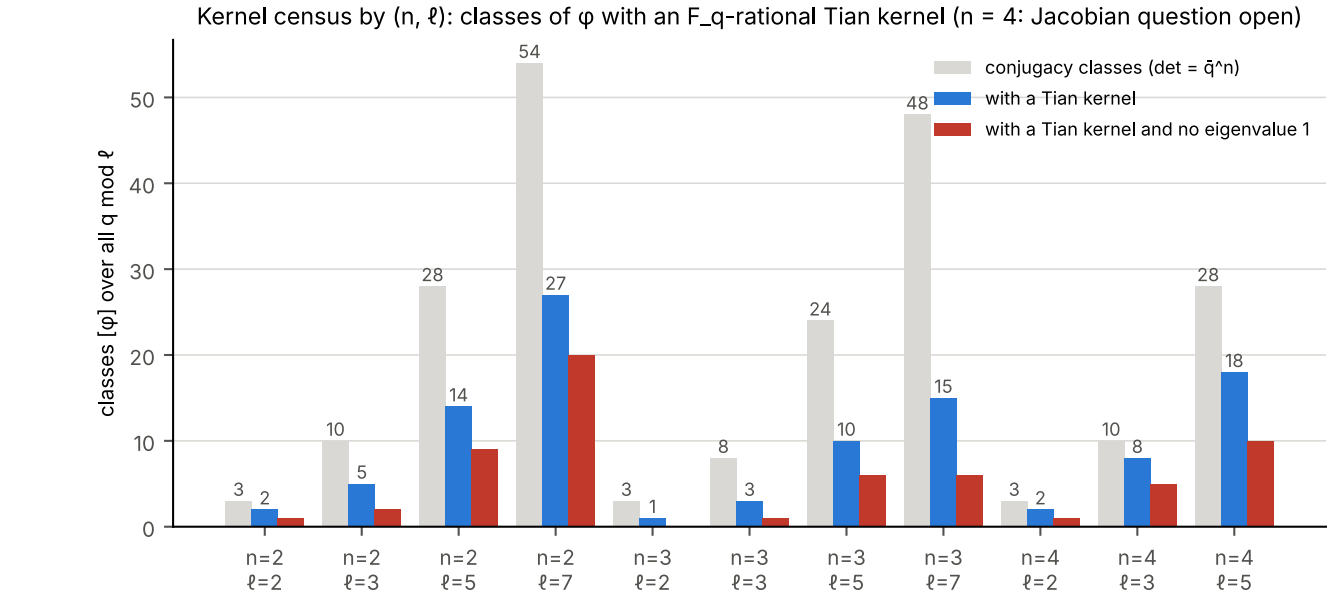
source: run-20261018T1605Z/summary.csv (exact counts)

Graph: n = 3: Tian kernels by class of φ



source: run-20261018T1605Z/kernels.csv (exact; one row per conjugacy class)

Graph: Kernel census by (n, ℓ)



source: run-20261018T1605Z/kernels.csv